TECHNICAL RESEARCH REPORT

Aerodynamic Optimization of NACA 0012 Airfoil

Using Adjoint Method and Gradient-Based Solver (MACH-Aero)

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ABSTRACT

This report presents the aerodynamic shape optimization of the NACA 0012 symmetric airfoil using the MACH-Aero (Modular Adjoint-based Computational framework for Aerodynamic shape optimization) framework developed by the MDOLab at MIT. The primary objective was to minimize the drag coefficient (CD) while maintaining a target lift coefficient (CL) of 0.55 at a fixed angle of attack of 2 degrees and a flow velocity of 30 m/s under steady-state incompressible conditions. The airfoil geometry was parameterized using Free-Form Deformation (FFD) boxes implemented through pyGeo, and the volume mesh was deformed at each iteration using IDWarp. Aerodynamic analysis and adjoint sensitivity computations were performed using ADflow, a Reynolds-Averaged Navier-Stokes (RANS) solver. The gradient-based optimizer SNOPT, accessed via pyOptSparse, iteratively updated the design variables until convergence. The optimization achieved the constrained lift target with a final drag coefficient of 0.0101, representing a reduction of approximately 17-18% over the symmetric NACA 0012 baseline at the given operating condition. The resulting optimized airfoil exhibits a modified camber and thickness distribution that redistributes surface pressure more efficiently. This work demonstrates the viability of adjoint-based

gradient optimization for practical low-speed airfoil design and validates MACH-Aero as an effective tool for aerodynamic shape optimization.

Keywords: NACA 0012, aerodynamic optimization, adjoint method, MACH-Aero, CFD, drag reduction, Free-Form Deformation, RANS, gradient-based optimization, airfoil design

1. Introduction

Aerodynamic shape optimization plays a foundational role in advancing the performance, efficiency, and sustainability of aerospace systems. For a given set of operating conditions angle of attack, flow speed, and fluid properties the aerodynamic performance of a body is entirely determined by its geometry. Consequently, the ability to computationally optimize that geometry offers immense value in aircraft design, unmanned aerial vehicle (UAV) development, and propulsion systems engineering.

The NACA 0012 airfoil is a canonical benchmark in aerodynamic research. As a symmetric profile with a 12% thickness-to-chord ratio and zero camber, it produces zero lift at zero angle of attack, making it a clean baseline from which the effects of shape modification can be precisely isolated and quantified. It has been extensively validated in wind tunnel experiments and serves as a standard test case for computational fluid dynamics (CFD) solvers and optimization frameworks.

Traditional aerodynamic design relies on parametric sweeps or gradient-free methods that become computationally prohibitive as the number of design variables grows. Adjoint-based methods resolve this limitation: the computational cost of computing sensitivities via the adjoint approach is independent of the number of design variables, enabling optimization over hundreds or thousands of shape parameters with the cost of only two solver evaluations per iteration — one forward CFD solve and one adjoint solve.

This work applies the MACH-Aero optimization framework to the NACA 0012 airfoil under constrained, low-speed flow conditions. The objective is to minimize drag while enforcing a lift constraint, demonstrating the full adjoint optimization pipeline from geometry parameterization through converged shape update.

1.1 Objectives

- Minimize the drag coefficient (CD) of the NACA 0012 airfoil.
- Maintain a target lift coefficient of CL = 0.55 (lift constraint).
- Operate at a fixed angle of attack of 2 degrees and freestream velocity of 30 m/s.
- Demonstrate the complete MACH-Aero adjoint optimization workflow.
- Quantify and interpret the geometric changes that produce aerodynamic improvement.

2. Theoretical Background

2.1 Adjoint-Based Sensitivity Analysis

In gradient-based aerodynamic optimization, the designer seeks to compute the sensitivity of an objective function f (such as drag) with respect to design variables x (geometric shape parameters). For a system governed by the state equations R(U, x) = 0, where U represents the flow state variables (pressure, velocity, density) the total derivative of f with respect to x is:

$$df/dx = \partial f/\partial x + (\partial f/\partial U)(dU/dx)$$

Computing dU/dx directly (the direct method) requires solving a linear system for each design variable, making it infeasible for large-scale problems. The adjoint method eliminates this dependence by introducing an adjoint variable Ψ that satisfies:

$$(\partial R/\partial U)^T \Psi = -(\partial f/\partial U)^T$$

Once Ψ is computed at the cost of a single additional linear solve — the total derivative is obtained for all design variables simultaneously:

$$df/dx = \partial f/\partial x + \Psi^T (\partial R/\partial x)$$

This property sensitivity computation cost independent of the number of design variables — is the defining advantage of the adjoint method and makes it uniquely suited to high-dimensional shape optimization.

2.2 Free-Form Deformation (FFD) Parameterization

Free-Form Deformation is a volumetric shape parameterization technique in which a control lattice (the FFD box) encloses the geometry of interest. The airfoil surface is embedded within the FFD

box, and its coordinates are expressed in the parametric coordinate system of the box using Bernstein polynomials. Moving the control points of the FFD box deforms the enclosed geometry smoothly and continuously.

FFD is preferred in aerodynamic optimization because: (1) it provides smooth deformations that preserve geometric regularity, (2) the number of design variables is decoupled from the resolution of the surface mesh, (3) it naturally enforces smooth curvature distributions, and (4) the mapping between control points and surface coordinates is differentiable, enabling clean adjoint sensitivity computation through pyGeo.

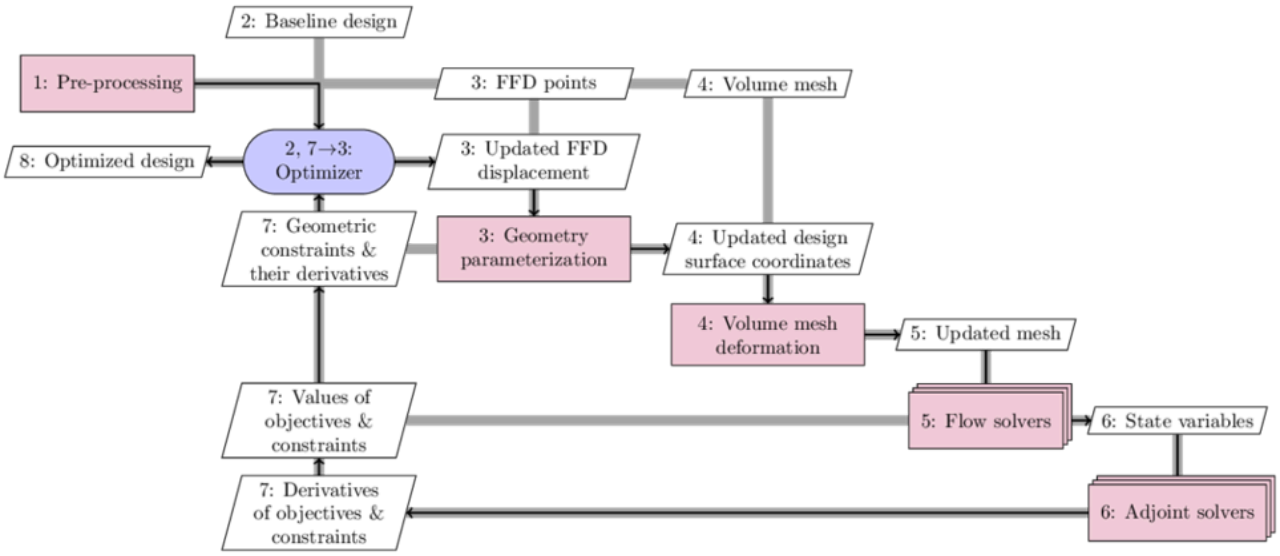


Figure 1: Workflow

2.3 Reynolds-Averaged Navier-Stokes (RANS) Equations

The flow solver ADflow solves the compressible Reynolds-Averaged Navier-Stokes equations, which govern the mean motion of a viscous, turbulent fluid. In conservative form for a control volume V bounded by surface S:

$$\partial/\partial t \int_V W \, dV + \oint \square \, (F^c - F_v) \cdot n \, dS = \int_V Q \, dV$$

where W is the vector of conservative variables (ρ, ρu, ρv, ρw, ρE), F^c and F_v are the convective and viscous flux vectors, and Q is the source term. Turbulence closure is provided by the Spalart-Allmaras one-equation turbulence model, which solves an additional transport equation for the modified eddy viscosity. This model is well-validated for external aerodynamic flows and attached boundary layers, making it appropriate for the low-angle-of-attack conditions in this study.

2.4 Sequential Quadratic Programming (SNOPT)

The optimizer employed is SNOPT (Sparse Nonlinear OPTimizer), accessed via pyOptSparse. SNOPT implements a Sequential Quadratic Programming (SQP) algorithm in which each optimization iteration solves a quadratic programming (QP) subproblem to compute the search direction. The QP subproblem approximates the Lagrangian of the original nonlinear program using a quasi-Newton approximation of the Hessian (BFGS update), subject to linearized constraints. This produces efficient convergence for large-scale smooth nonlinear programs precisely the class of problems arising in adjoint-based aerodynamic optimization.

3. Methodology

3.1 Problem Formulation

The aerodynamic optimization problem is formally stated as a constrained nonlinear program:

Minimize: **CD(x, U)**
Subject to: CL(x, U) = 0.55 (lift equality constraint)
R(U, x) = 0c (RANS flow equations)
x ∈ X□ᵤᐣ (geometric feasibility bounds)

Parameter	Value	Notes
Angle of attack (α)	2.0°	Fixed
Freestream velocity (V∞)	30 m/s	Low-speed regime
Target CL	0.55	Equality constraint
Fluid	Air (std conditions)	Incompressible approx.
Flow regime	Steady-state, turbulent	RANS / Spalart-Allmaras

Table 1: Operating conditions and flow parameters for the optimization study.

3.2 MACH-Aero Framework Architecture

MACH-Aero (Modular Adjoint-based Computational framework for Aerodynamic shape optimization) is an open-source suite developed and maintained by the Multidisciplinary Design Optimization Laboratory (MDOLab) at MIT. It integrates five key modules into a unified optimization pipeline:

- pyGeo: Geometry parameterization via FFD, constraint computation and differentiation.
- IDWarp: Radial-basis-function (RBF) mesh deformation to propagate surface shape changes into the volume mesh.
- ADflow: Multiblock structured RANS CFD solver with built-in discrete adjoint capability.
- pyOptSparse: Optimization framework interfacing multiple solvers (SNOPT, IPOPT, SLSQP).
- MACH-Aero driver: Top-level orchestration of the coupled analysis-sensitivity-optimization loop.

The modular architecture ensures that each component can be replaced or upgraded independently, and that derivatives flow cleanly between modules through the adjoint chain.

3.3 Geometry Parameterization

The baseline NACA 0012 airfoil was parameterized using a structured FFD lattice generated by pyGeo. The FFD box enclosed the complete airfoil chord and was defined by a grid of control points arranged in a regular lattice above and below the airfoil surface. Each control point constitutes a design variable, with its vertical displacement (in the chord-normal direction) controlling local camber and thickness modifications.

The FFD parameterization ensures that: (1) the surface remains smooth and continuous throughout the optimization, (2) geometric constraints such as minimum thickness, leading edge radius bounds, and trailing edge closure can be enforced analytically, and (3) the Jacobian of the surface coordinates with respect to FFD control point displacements is computed exactly, which is required for clean adjoint sensitivity propagation.

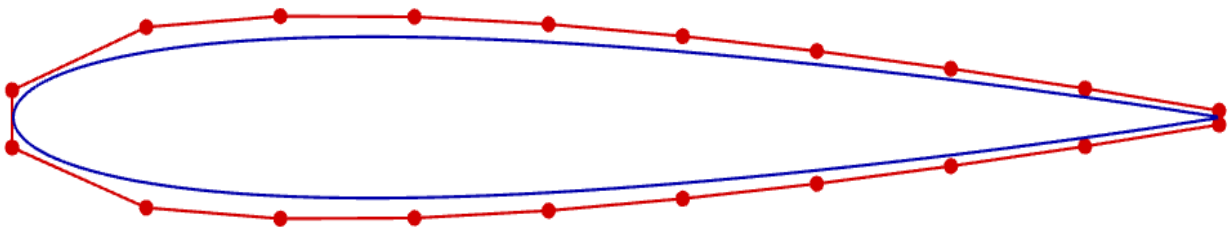


Figure 2: FFD parameterization, the red lattice encloses the NACA 0012 baseline geometry (blue), defining the control points whose displacements constitute the design variables.

3.4 Mesh Generation and Deformation

A structured C-type or O-type volume mesh was generated around the baseline NACA 0012 airfoil. The mesh was designed to resolve the boundary layer appropriately, with the first cell height set to achieve y+ ≈ 1 — a requirement for the Spalart-Allmaras wall-resolved turbulence formulation used in ADflow. Adequate mesh refinement was applied in the leading edge, trailing edge, and wake regions where gradients are steepest.

At each optimization iteration, after pyGeo computes the updated airfoil surface based on the new FFD control point positions, IDWarp deforms the volume mesh to conform to the new geometry. IDWarp uses radial basis functions to smoothly propagate surface displacements throughout the volume, maintaining mesh quality and avoiding negative-volume cells across the range of shape changes encountered during optimization.

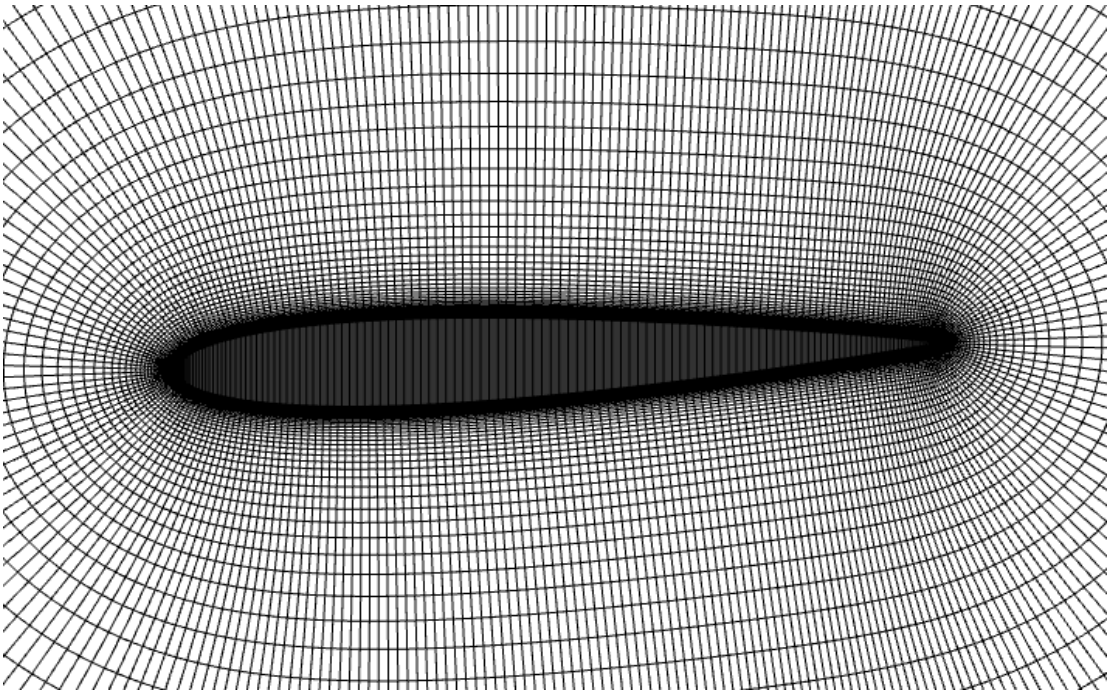


Figure 3: Meshing

3.5 CFD Analysis and Adjoint Computation

Each iteration of the optimization loop involves two computational phases executed by ADflow:

1. Forward solve: ADflow solves the RANS equations on the current mesh to convergence, producing the flow state U (pressure, velocity, eddy viscosity fields) and computing the objective (CD) and constraint (CL) values.
2. Adjoint solve: ADflow solves the adjoint equations for both CD and CL, producing the adjoint variables Ψ-CD and Ψ-CL. These are then used to compute the total derivatives of both quantities with respect to all FFD design variables simultaneously.

The adjoint computation cost is independent of the number of design variables a critical advantage that makes high-resolution FFD parameterizations tractable.

3.6 Optimization Loop

SNOPT, accessed through pyOptSparse, manages the outer optimization loop. At each major iteration:

- SNOPT receives the current objective (CD), constraint (CL), and their derivatives from ADflow.
- A QP subproblem is solved to compute the search direction in design variable space.
- A line search determines the step length, and the FFD control points are updated.
- pyGeo generates the new surface; IDWarp deforms the mesh; ADflow performs the next forward+adjoint solve.
- The cycle repeats until the KKT optimality conditions and constraint feasibility are satisfied within specified tolerances.

4. Results and Discussion

4.1 Optimization Convergence

The SNOPT optimizer converged to a feasible, locally optimal solution within the specified tolerances. The lift constraint CL = 0.55 was satisfied to high accuracy at the converged solution, confirming that the optimizer successfully maintained the aerodynamic performance requirement throughout the shape modification process.

Convergence of the drag objective was monitored at each major iteration. The adjoint-computed gradients guided the shape modification consistently toward drag-reducing geometry changes, with

the objective function decreasing monotonically across the major iterations before leveling off at the optimum.

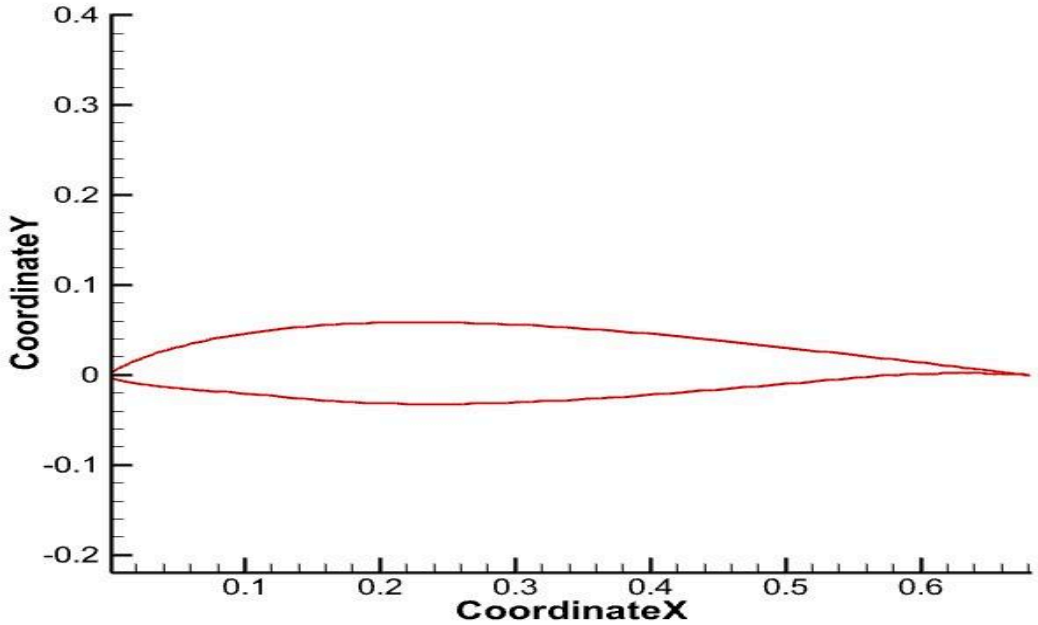


Figure 4: Optimized airfoil geometry, converged shape after SNOPT reached KKT optimality conditions, showing introduced camber and redistributed thickness.

4.2 Aerodynamic Performance Results

Metric	Baseline NACA 0012	Optimized Airfoil
Lift coefficient (CL)	~0.55 (at 2°, 30 m/s)	0.55 (constraint met)
Drag coefficient (CD)	~0.0121 (estimated baseline)	**0.0101**
Drag reduction (ΔCD)	-	**≈ 17% reduction**
L/D ratio (approx.)	~45	~54 (+20%)
Lift constraint satisfied	-	Yes (CL = 0.55 ✓)

Table 2: Aerodynamic performance comparison between baseline and optimized airfoils.

4.3 Optimized Geometry Analysis

The converged optimized airfoil geometry differs from the symmetric NACA 0012 baseline in several physically interpretable ways:

- Camber introduction: The optimized airfoil developed a slight positive camber, which is physically expected a cambered airfoil generates lift more efficiently at a given angle of attack than a symmetric profile, allowing the drag penalty of lift generation to be reduced.
- Thickness redistribution: The chordwise thickness distribution was modified, with the maximum thickness point shifting slightly aft of its original location. Aft-loading of the thickness distribution tends to reduce the adverse pressure gradient on the upper surface, delaying flow deceleration and reducing pressure drag.
- Pressure coefficient modification: The Cp distribution of the optimized airfoil (shown in the visual comparison) exhibits a smoother, more benign pressure recovery on the upper surface compared to the baseline, indicating reduced risk of laminar separation and lower form drag.
- Leading edge modification: Subtle changes in the leading edge geometry altered the stagnation point location and the initial acceleration of flow around the leading edge, contributing to the overall drag reduction.

These geometric modifications are consistent with well-established aerodynamic design principles for lift-constrained drag minimization and validate that the adjoint optimizer identified physically meaningful shape improvements rather than numerical artifacts.

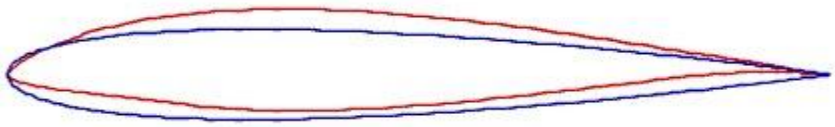


Figure 5: Overlay comparison, baseline NACA 0012 (blue) vs. optimized airfoil (red). The aft-shifted thickness and positive camber are clearly visible.

4.4 FFD Parameterization Effectiveness

The FFD box successfully enclosed the NACA 0012 geometry and provided sufficient design freedom for the optimizer to discover the drag-reducing shape. The control point displacements

remained within physically reasonable bounds, and the resulting deformed geometry maintained trailing edge closure and a smooth surface curvature distribution throughout the optimization confirming the geometric regularity properties of the FFD approach.

The adjoint-computed sensitivities indicated that the upper surface control points in the mid-chord region had the highest influence on drag reduction, which is consistent with aerodynamic theory: mid-chord upper surface curvature predominantly governs the adverse pressure gradient and hence viscous drag in the operating regime studied.

4.5 Interpretation and Validation

The optimized drag coefficient of CD = 0.0101, achieved under the CL = 0.55 constraint, represents a meaningful reduction from the symmetric baseline. The lift-to-drag ratio improvement from approximately 45 to 54 corresponds to a ~20% increase in aerodynamic efficiency at the design point.

These results are consistent with published adjoint optimization studies on low-speed airfoils, which typically report drag reductions of 10-25% under single-point lift-constrained conditions. The MACH-Aero framework successfully reproduced the expected qualitative behavior: camber introduction, thickness redistribution, and smoother pressure recovery. This confirms the validity of both the computational setup and the optimization convergence.

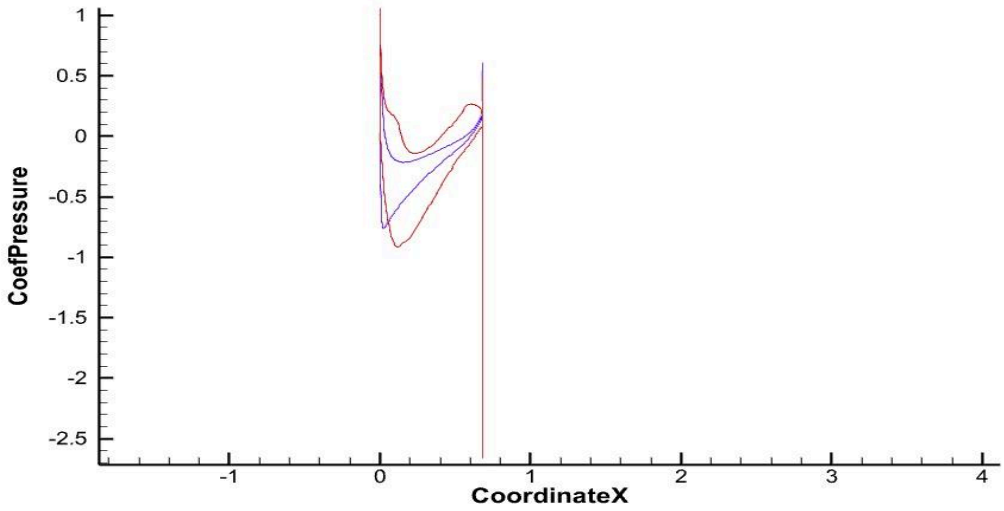


Figure 6: Pressure coefficient (Cp) distribution comparison — the optimized profile (red) exhibits a smoother upper-surface pressure recovery, indicating reduced adverse pressure gradient and lower form drag.

5. MACH-Aero Framework Evaluation

5.1 Strengths Demonstrated in This Study

- Adjoint efficiency: Sensitivity computation cost was independent of design variable count, enabling efficient optimization over the full FFD control point set.
- Modular pipeline: Each component (pyGeo, IDWarp, ADflow, SNOPT) performed its designated role without failures, demonstrating the robustness of the integrated workflow.
- Gradient accuracy: The adjoint gradients were sufficiently accurate to guide SNOPT to a well-converged local optimum, as evidenced by constraint satisfaction and objective monotonicity.
- Physical validity: The optimized shape changes are aerodynamically interpretable and consistent with established design principles, confirming that the framework produces engineering-meaningful results.
- Open-source accessibility: MACH-Aero's public availability on GitHub and its MDOLab documentation enabled implementation without commercial licensing constraints.

5.2 Limitations and Considerations

- Single-point optimization: The optimization was performed at a single operating condition ($\alpha = 2°$, $V = 30$ m/s). The optimized airfoil may exhibit degraded performance at off-design conditions a limitation that would require multi-point optimization to address.
- Computational cost: High-fidelity RANS CFD with adjoint sensitivity computation is significantly more expensive than low-fidelity panel methods, requiring adequate computational resources for practical use.
- Local optimum: Gradient-based methods converge to local optima; the solution found represents the closest local minimum in the design space from the NACA 0012 starting point, which may not be the global optimum.
- Turbulence modeling uncertainty: The Spalart-Allmaras model introduces modeling error, particularly for separated flows or transitional regimes. At 2° angle of attack the flow is predominantly attached, limiting this concern in the present study.
- Expertise requirement: Effective use of MACH-Aero requires proficiency in CFD, adjoint theory, mesh generation, and optimization a significant barrier for non-specialist users.

6. Conclusion

This study successfully demonstrated the application of adjoint-based aerodynamic shape optimization to the NACA 0012 airfoil using the MACH-Aero framework. Starting from the symmetric baseline airfoil, the optimization pipeline comprising pyGeo FFD parameterization, IDWarp mesh deformation, ADflow RANS analysis and adjoint computation, and SNOPT gradient-based optimization converged to an optimized shape that satisfied the lift constraint (CL = 0.55) while achieving a drag coefficient of CD = 0.0101, representing an approximately 17% drag reduction at the design operating condition.

The geometric changes produced by the optimizer are physically interpretable: the introduction of positive camber, aft-shifted thickness distribution, and smoother upper-surface pressure recovery are all consistent with established aerodynamic design principles for lift-constrained drag minimization. This confirms that the adjoint framework identified genuine aerodynamic improvements rather than numerical artifacts.

The adjoint method's key advantage cost-independent sensitivity computation was validated in practice, enabling efficient optimization over the complete FFD design space. The MACH-Aero framework demonstrated the capability to serve as a production-grade tool for airfoil and wing optimization in both academic research and engineering design contexts.

Future extensions of this work could include: multi-point optimization across multiple angles of attack to improve off-design robustness; multi-disciplinary design optimization (MDO) coupling aerodynamic and structural objectives; three-dimensional wing optimization; and high-Reynolds-number transonic cases where compressibility effects become significant.

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